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SemanticDistance: An R package for Computing Semantic Distance in Structured Texts and Visualizing Clustering Properties in Unstructured Word Lists

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Abstract

`SemanticDistance` computes pairwise distance between constituents (e.g., word-to-word, ngram-to-word, turn-to-turn) in both structured language samples and unstructured word lists. `SemanticDistance` has cleaning and formatting options including stopword removal and lemmatization. The package computes two complementary cosine distance indices for each pairwise contrast of interest. `SemanticDistance` can also be used to examine clustering properties within unstructured word lists, generating dendrograms and simple igraph network plots.

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Statment of Need

`SemanticDistance` is unique in its capacity to compute semantic distance metrics within running language samples (e.g., word-to-word, bigram-to-word) and to produce semantic network visualizations. This user-friendly application will allow researchers to model the flow of meaning in naturalistic language samples.

Description

There are many ways to assess similarity between concepts (Malt, 2020). Consider, for example, a wildlife biologist interested in quantifying distance between *wolf* and *dog* in terms of perceived threat to humans. How would she do this? One approach might involve asking people to explicitly rate perceived threat of dogs and wolves. The difference in threat ratings (dog vs. wolf) represents a simple form of semantic distance in a one-dimensional semantic space constrained by threat (for higher dimensional approaches see Pennington et al. (2014)).

`SemanticDistance` computes two complementary distance metrics between pairs of constituents (e.g., words, ngrams, turns). Distance values are derived from two custom lookup databases containing semantic vectors for >70k English words. `CosDist_Glo` reflects cosine distance between vectors derived from training a GLOVE word embedding model (Pennington et al., 2014), whereas `CosDist_SD15` refects cosine distances derived from a 15 dimension sensorimotor and affective space (Reilly et al., 2023). `SemanticDistance` processes distance relationships between lexical constituents within the following text formats:

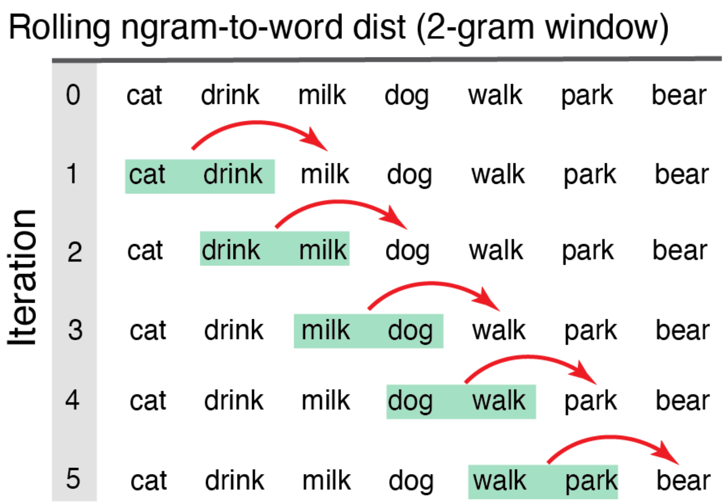
1: **monologues**: stories, narratives, and structured text
2: **dialogues**: two-person conversation transcripts.
3: **word pairs in columns**: word pairs
4: **unordered lists**: bags-of-words

`SemanticDistance` bundles text cleaning and formatting options (e.g., stopword removal, lemmatization) with distance and network visualization options. `SemanticDistance` also generates rolling measures of semantic distance within structured text. In monologues (stories, narratives) and dialogues (two-person conversation transcripts), word order matters. Unordered word lists (e.g., a grocery list) are different and can be thought of as non-continuous bags-of-words. `SemanticDistance` can run distance metrics on both of these formats with options for chunk size guided by the user.

Examples of Distance Functions and Possible Applications

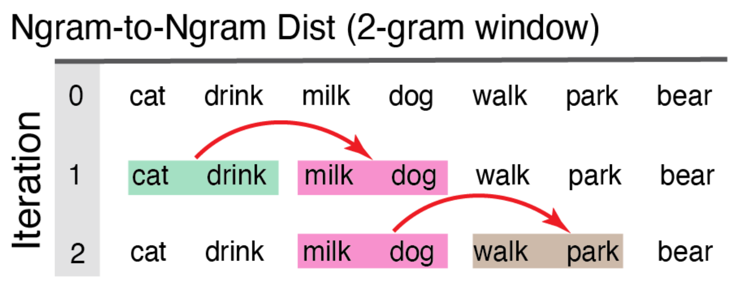
Ngram-to-Word Distance

Computes rolling distance across ordered language samples (see Figure 1). This metric may be used to examine larger chunks from each word to its prior context.



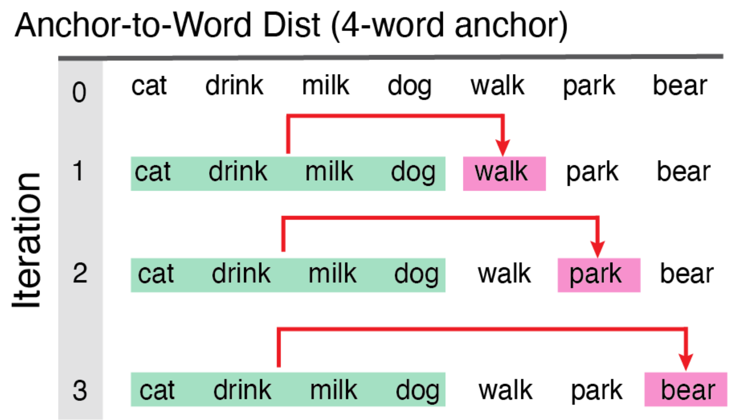
Ngram-to-Ngram Distance (monologues, stories, continuous structured language)

Computes a rolling measure of semantic distance ngram-to-ngram across an ordered language sample (see Figure 2). This metric may be used to examine semantic cohesion between smaller vs. larger chunks of words within linear narrative discourse.



Anchored Distance (monologues, stories, continuous structured language)

Computes semantic distance iteratively for each new word in an ordered language sample (e.g., story, narrative) relative to the initial block of n content words (see Figure 3). Anchored distance can provide an empirical measure of semantic drift or topic variability over the course of a sample (e.g., started with sports, jumped to baking, then to politics).



Turn-by-Turn Distance (2-person Dyads, Conversation Tramscripts)

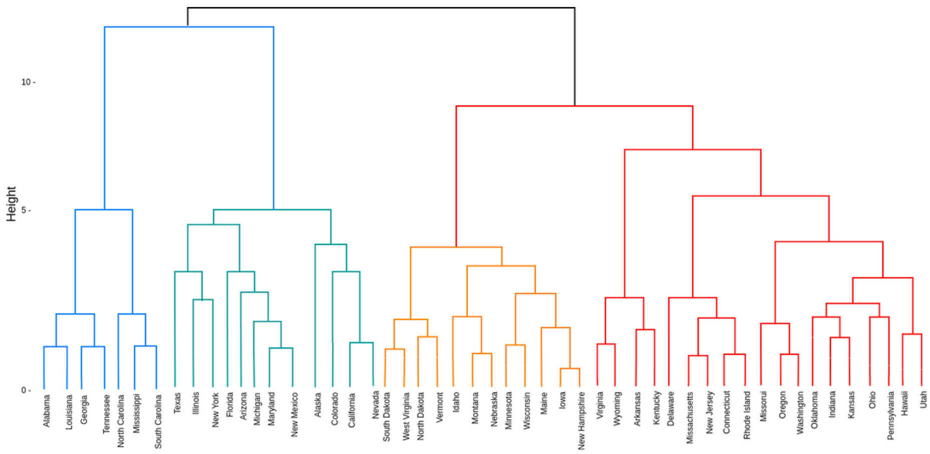
Computes a rolling measure of semantic distance turn-to-turn in a conversation/dialogue transcript. Semantic vectors for all content words within one turn are averaged and grouped by speaker identity. The function then computes cosine distance between each speaker at each turn, yielding a fluctuating time series of distance changes by speaker. This measure could be used to assess lexical-semantic comprehension and alignment in naturalistic conversations.

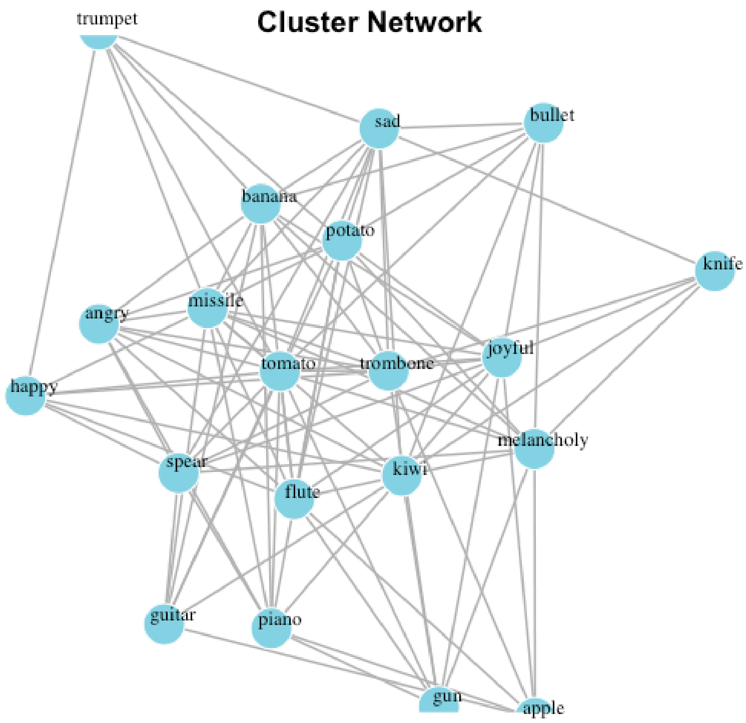
Word Pair Distance (user arrayed word pairs in columns)

Computes pairwise distance metrics row-wise between two specified target columns in a dataframe, This function is useful for stimulus norming or deriving distances for a specified set of non-continuous word pairs.

Finding structure in the chaos of an unstructured word list

`SemanticDistance` includes visualization options. The `wordlist_to_network` function takes a word list as an argument and derives cluster dendrograms (see Figure 4) or network plots (see Figure 5).





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